AARTI STEEL INTERNATIONAL LIMITED

Ludhiana, Punjab

INDUSTRIAL INTERNSHIP REPORT

Complete Steel Making Technical Report

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CHAPTER 1

SCRAP HANDLING

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ELECTRIC ARC FURNACE

1. Introduction and Overall Steelmaking Route

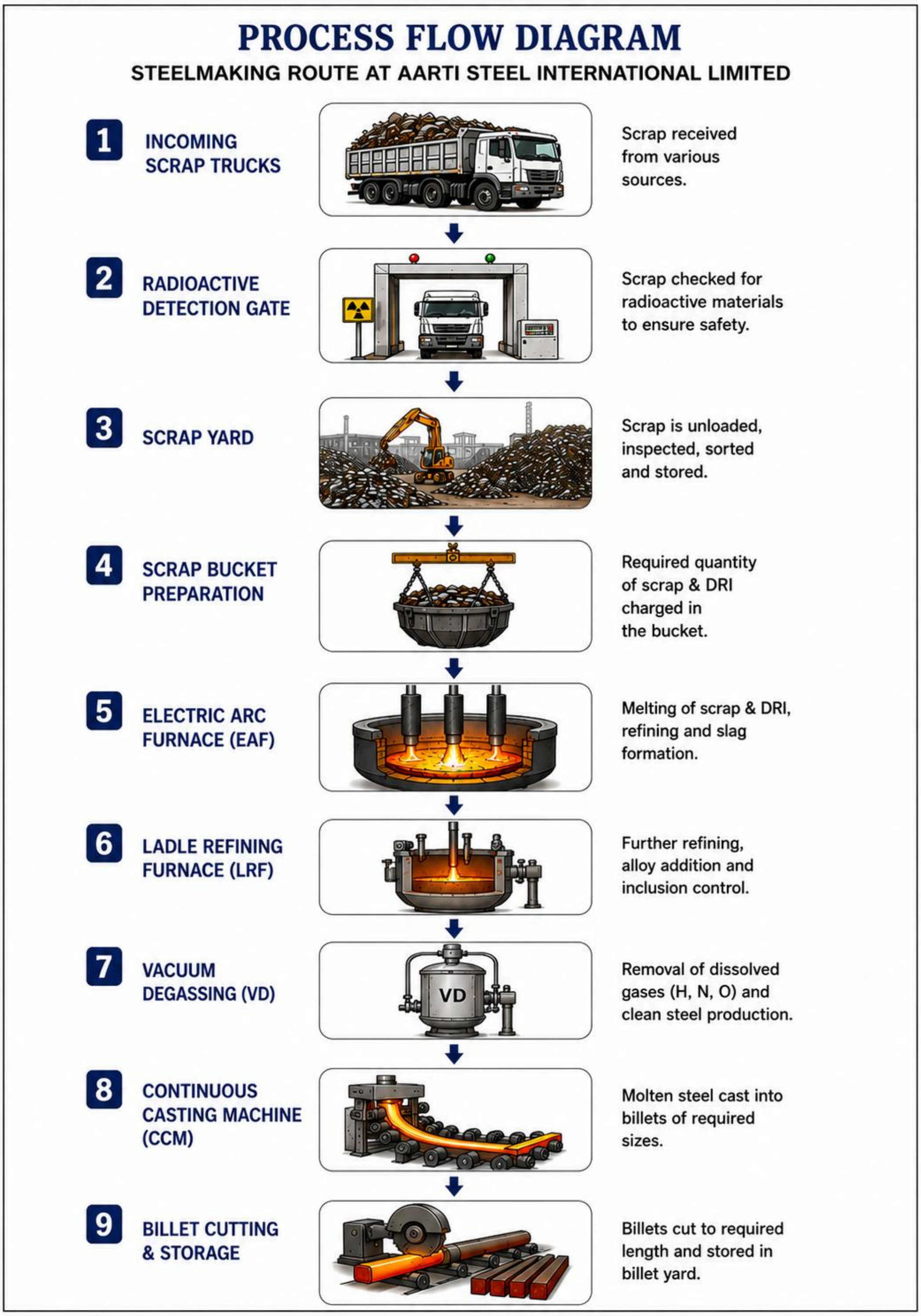

Figure 1.1: *Overall Steelmaking route at Aarti Steel International Limited*

Steel production at Aarti Steel International Limited (ASIL), Ludhiana, follows the Electric Arc Furnace (EAF) route. In this process, steel scrap and Direct Reduced Iron (DRI) are melted and refined to produce high-quality liquid steel. The EAF route offers significant advantages over conventional ironmaking processes due to its flexibility, lower capital investment, shorter production cycle, and ability to recycle large quantities of steel scrap.

The complete steelmaking route at ASIL begins with the inspection and classification of incoming scrap and ends with the production of billets through the Continuous Casting Machine (CCM). Throughout this route, various metallurgical operations are carried out to remove impurities, adjust chemical composition, and achieve the desired steel quality. The EAF acts as the primary melting unit where electrical and chemical energy are used to convert solid metallic charge materials into molten steel. The molten steel is subsequently refined in the LRF and VD before being cast into billets in the CCM.

2. Scrap Handling

2.1 Introduction to Steel Scrap

Steel scrap is the principal raw material used in EAF steelmaking. It consists of previously manufactured steel products that have reached the end of their service life, as well as steel generated during industrial manufacturing operations. The use of scrap significantly reduces the requirement for virgin ironmaking, contributing to resource conservation and environmental sustainability. However, variations in scrap chemistry, density, cleanliness, and residual element content can strongly influence furnace productivity and steel quality. Therefore, proper inspection, classification, and preparation are essential before charging.

2.2 Radioactive Detection Gate

Figure 1.2: *Radioactive detection gate*

Before entering the steel plant premises, every scrap-carrying vehicle passes through a radioactive monitoring system. The primary purpose of this system is to ensure that no radioactive material enters the steelmaking process contamination can originate from discarded medical equipment, industrial gauges, or improperly disposed radioactive sources. If not detected, such contamination can result in:

- Contamination of liquid steel and furnace refractories
- Damage to dust collection systems and plant equipment
- Severe environmental, health, and economic impacts

The gate operates by positioning highly sensitive radiation detectors on both sides of the truck path to continuously monitor gamma radiation. If radiation levels exceed the permissible limit, an alarm activates and the vehicle is isolated for inspection. This system serves to:

- Prevent contamination of steel products
- Protect plant personnel from radiation exposure
- Ensure regulatory compliance
- Avoid expensive decontamination procedures

2.3 Scrap Inspection, Classification, and Types

After clearing the radioactive gate, the scrap is unloaded in the scrap yard and visually inspected. Proper classification ensures predictable furnace performance and helps achieve the desired steel chemistry. The inspection process involves:

- Identification of scrap type and grade
- Removal of non-metallic contaminants
- Detection of oil, grease, paint, rubber, and plastic
- Identification of oversized pieces requiring cutting
- Sorting according to density and dimensions

Several types of metallic charge materials are used at ASIL. The table below summarises each type along with its source and role in the EAF:

Scrap Type	Source	Role in EAF
Heavy Melting Scrap (HMS)	Demolished structures, heavy machinery	Base charge; high density, good metallic recovery
Wire bundles	Discarded electrical wires and cables	To increase copper content
Plate & Structural Scrap	Steel plates, channels, beams	High metallic content, stable chemistry
Turnings & Borings	Machining and drilling operations	Void filling; improves bucket packing efficiency
Cast Iron Scrap	Machine parts and castings	High carbon source; improves metallic yield
Pig Iron	Blast furnace output (3.5–4.5% C)	Extra carbon and iron units
Direct Reduced Iron (DRI)	Solid-state reduction of iron ore (~10–12 t/heat)	Clean iron units; low residuals; improves cleanliness


(a)
(b)
(c)
(d)

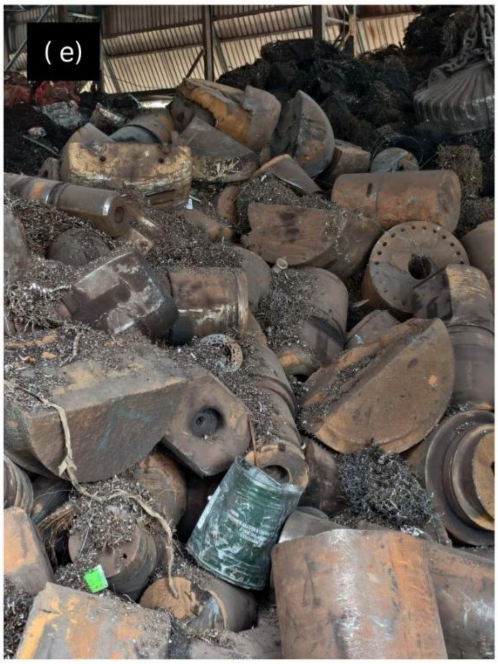

Figure 1.3: *Types of metallic charge materials used in the Electric Arc Furnace at Aarti Steel International Limited (a) Heavy Melting Scrap, (b) Wire Bundles, (c) Plate & Structural Scrap, (d) Turnings & Borings, (e) Cast Iron Scrap, (f) Pig Iron.*

3. Electric Arc Furnace: Principles and Design

3.1 Introduction and Advantages

The Electric Arc Furnace (EAF) is the primary melting unit in the steel melting shop of ASIL. The furnace converts electrical energy into thermal energy through electric arcs generated between graphite electrodes and the metallic charge. The EAF installed at ASIL has a capacity of approximately 35 tonnes per heat and utilises a combination of electrical energy, oxygen injection, carbon injection, and slag engineering to efficiently produce molten steel. It acts as a high-temperature metallurgical reactor capable of reaching over 1600°C, performing melting, oxidation, dephosphorization, and preliminary refining.

Compared to conventional Blast Furnace routes, the EAF offers the following advantages:

- High scrap utilisation and flexible production
- Lower environmental impact
- Faster startup and shutdown times
- Lower capital investment
- Better adaptability to multiple steel grades

3.2 Principle of Arc Generation and Heat Transfer

The working principle of an EAF is based on generating an electric arc between graphite electrodes and the metallic charge. When a sufficiently high voltage is applied, dielectric breakdown of the surrounding air occurs and the air becomes ionised, transforming into a conductive plasma channel. Once ionisation begins, electric current flows through the plasma, generating an arc whose temperature may exceed 3000–5000°C. This intense thermal radiation rapidly melts the scrap and initiates steelmaking reactions.

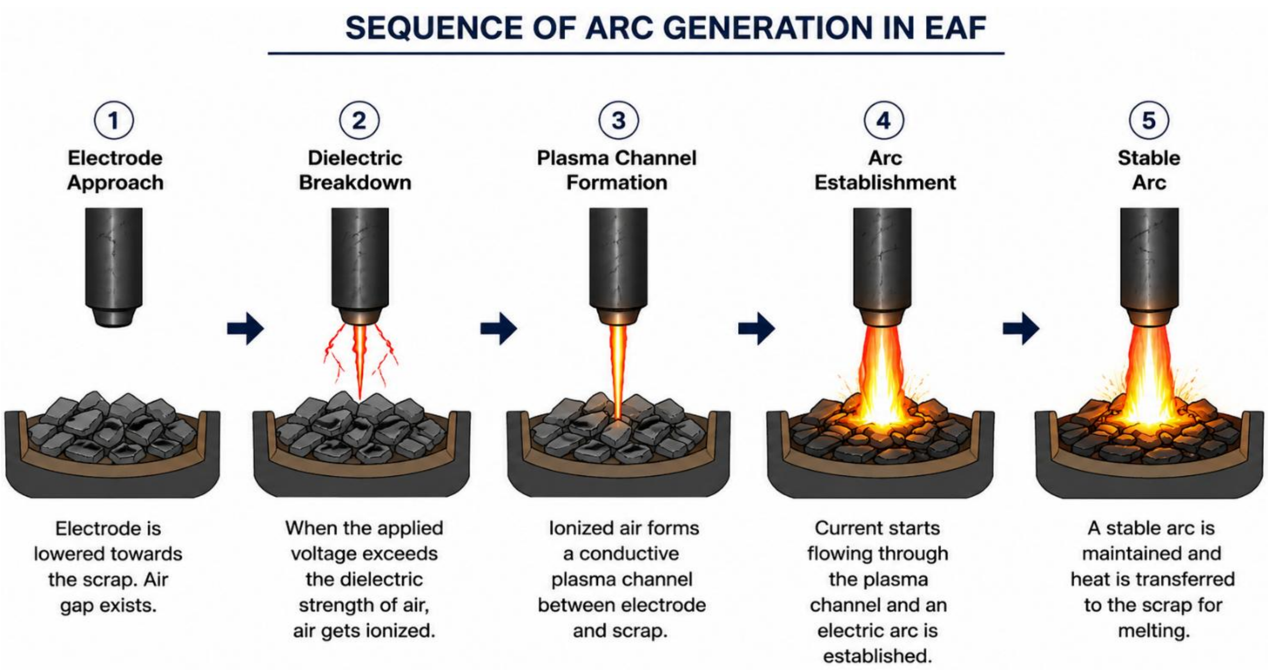

Figure 1.4: *Sequence of arc generation in EAF*

Three main mechanisms contribute to heating within the furnace:

- **Radiation:** the dominant mode, with approximately 70–80% of arc energy reaching the scrap through thermal radiation
- **Conduction:** transfers heat from hot molten steel to colder regions within the bath
- **Convection:** currents circulate molten steel within the bath, promoting temperature and composition uniformity

3.3 Construction and Major Components

The EAF at ASIL consists of several mechanical, electrical, hydraulic, and metallurgical systems that operate together to ensure efficient steel production.

Furnace Shell

The furnace shell is the outer steel structure supporting the refractory lining and all furnace equipment. It is divided into two regions, the Lower Shell, which contains the hearth refractory, molten steel bath, and slag layer; and the Upper Shell, which contains water-cooled panels, sidewall refractories, and the slag door.

Furnace Hearth

The hearth is the bottom portion of the furnace where molten steel accumulates during operation. It is lined with basic refractories such as MgO-C bricks, Magnesia bricks, and Doloma bricks, owing to the highly basic nature of steelmaking slags. It experiences severe chemical attack, thermal shock, and mechanical erosion, making regular maintenance necessary.

Sidewalls

The sidewalls form the vertical section of the furnace, containing molten steel and slag while protecting the shell and supporting the roof. They are exposed to arc radiation, slag attack, metal splashing, and oxygen jet impingement, making refractory wear in this zone generally higher than in many other furnace regions.

Furnace Roof

The furnace roof closes the furnace during operation and supports the graphite electrodes. It contains electrode openings, water cooling systems, and fume extraction arrangements. It is hydraulically swung away during charging and returned before power-on, and uses extensive water cooling to improve service life.

Water Cooling Panels

Modern EAFs employ water-cooled panels in the upper shell and roof to manage extreme heat, continuously circulating water through copper or steel channels. This reduces refractory consumption, improves furnace life, minimises shell overheating, and increases overall productivity. Any interruption in cooling water flow can result in severe equipment damage and safety hazards.

Transformer, Bus Tubes, and Electrode Arms

The transformer is one of the most critical electrical components of the EAF. It converts high-voltage incoming power into low-voltage, high-current power suitable for arc generation, with tap settings adjusted throughout the heat to optimise arc stability, energy transfer, and melting rate. Current is carried from the transformer to the graphite electrodes through water-cooled copper bus tubes, while the electrode arms support and position the electrodes, transferring power to them with minimal electrical losses.

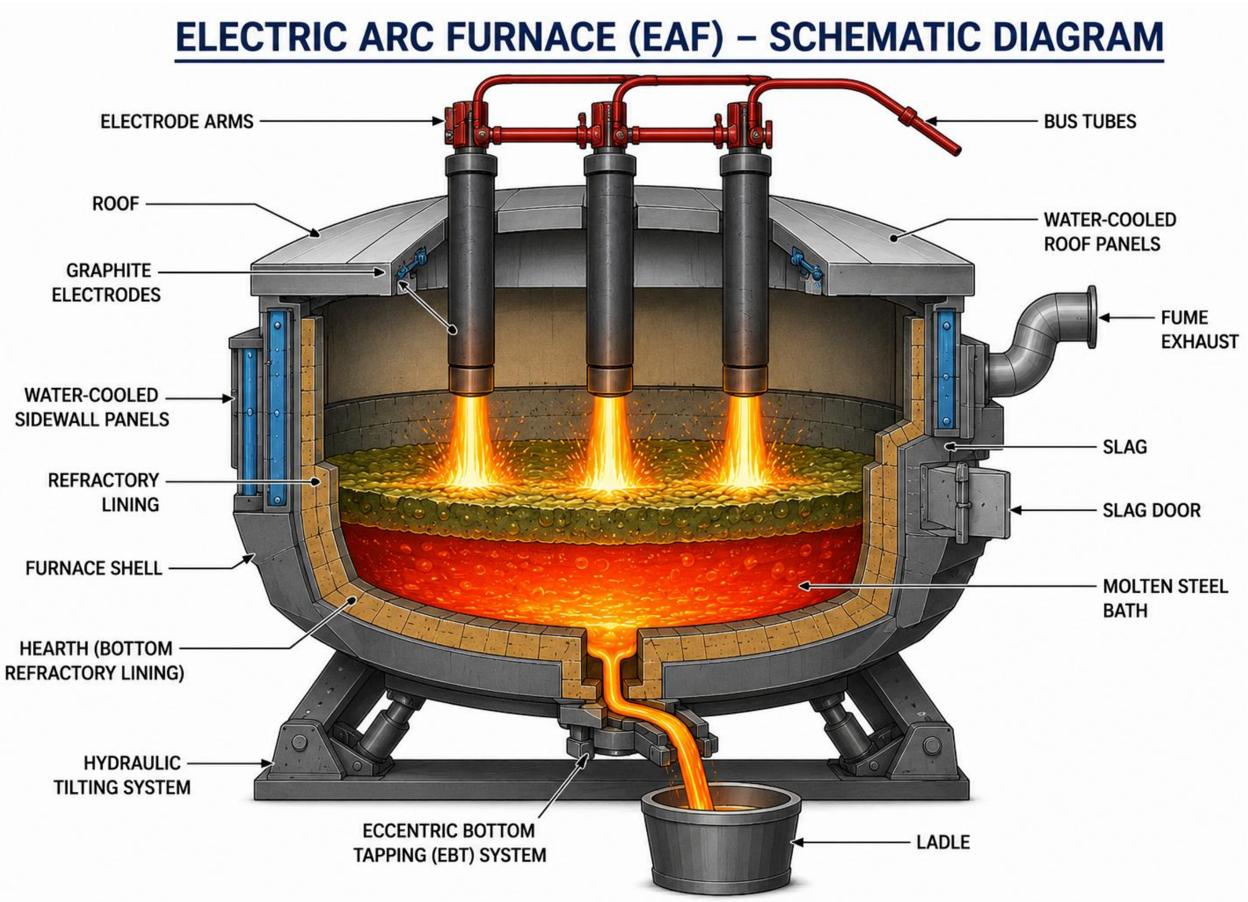

Figure 1.5: *EAF Schematic diagram*

4. Graphite Electrodes and AC vs. DC EAF

4.1 Role and Consumption of Graphite Electrodes

Graphite electrodes are responsible for generating the electric arcs required for melting scrap. The EAF at ASIL uses three graphite electrodes approximately 16 feet in length. They conduct enormous currents while maintaining structural stability under extreme thermal conditions and are continuously consumed during steelmaking, directly affecting furnace operating costs. The major consumption mechanisms include:

- **Oxidation:** carbon reacts with oxygen: $C + O_2 \rightarrow CO_2$
- **Side oxidation:** electrode side surfaces react with oxygen in the furnace atmosphere
- **Sublimation:** carbon vaporises at extremely high temperatures
- **Mechanical breakage:** due to improper handling, scrap collapse, or arc instability
- **Joint failure:** caused by poor tightening or excessive stress at threaded connections

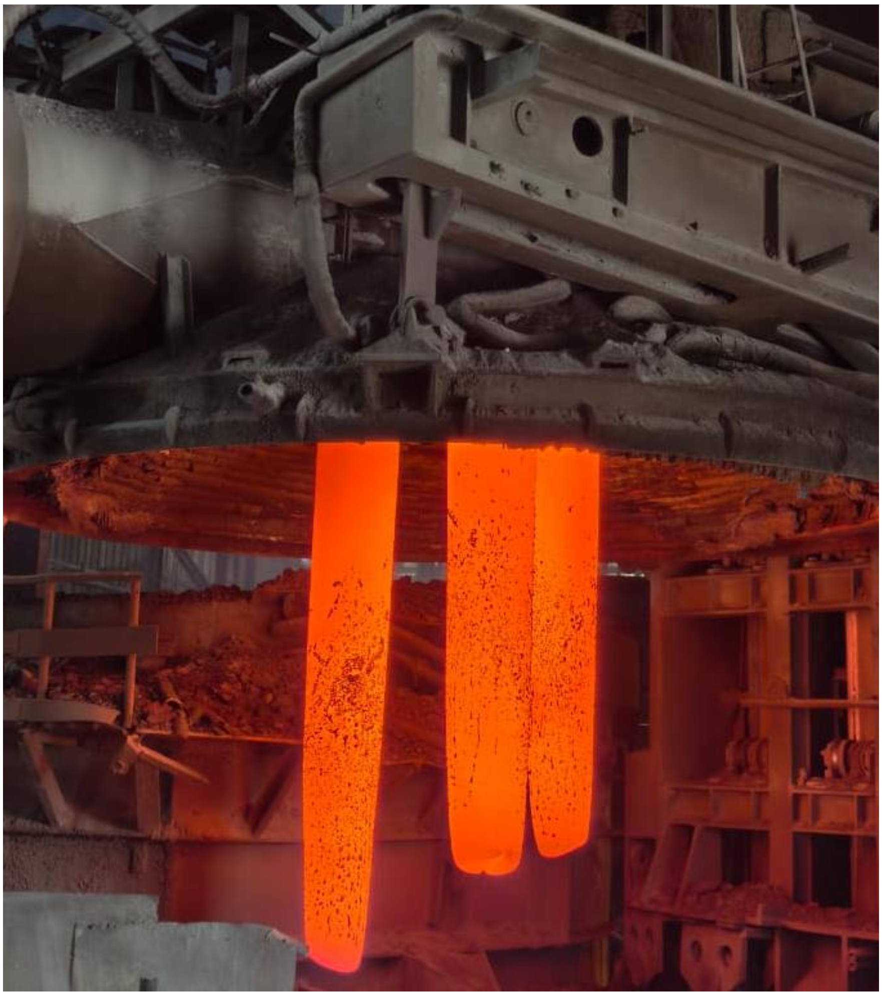

Figure1. 6: *3 Graphite electrodes of EAF*

4.2 AC EAF versus DC EAF

Because of its reliability and flexibility, the AC EAF remains the dominant technology in steelmaking plants worldwide. The table below compares the key features of AC and DC electric arc furnaces:

Feature	AC EAF	DC EAF
Electrodes	Three graphite electrodes	One top electrode + bottom anode
Power Supply	Three-phase AC supply	DC supply
Advantages	Simpler design, lower capital cost, easier maintenance, widely adopted	Lower electrode consumption, stable arc, lower electrical flicker
Disadvantages	Higher electrode consumption, greater electrical disturbances	Higher installation cost, complex bottom electrode system

5. Refractories and Gunning Mass

5.1 Importance and Zones of Refractories

Refractories protect the steel shell from direct exposure to molten steel and slag. They must resist temperatures above 1600°C, slag corrosion, thermal shock, and mechanical wear. The EAF refractory system is organised into three zones:

- **Working Lining:** directly contacts the molten steel and slag
- **Safety Lining:** provides backup protection in case the working lining fails
- **Permanent Lining:** the innermost layer protecting the furnace shell

5.2 Refractory Brick Types

MgO-C Bricks (Magnesia-Carbon) are the most commonly used EAF refractories, offering excellent slag resistance, high thermal shock resistance, and low wettability by slag. Their typical composition is as follows:

Component	Percentage
MgO (Magnesia)	80–90%
Carbon	10–20%

Doloma Bricks contain CaO and MgO, providing good resistance to basic slags and offering an excellent dephosphorization environment.

5.3 Gunning Mass

Gunning mass is used to repair worn refractory areas between heats. The material is pneumatically sprayed onto damaged regions showing excessive wear, extending refractory life, reducing downtime, and preventing localised failures. Its composition is given below:

Component	Value
LOI (Loss on Ignition)	4.00%
SiO_2	6.20%
MgO	84.67%

6. Eccentric Bottom Tapping (EBT)

The EAF at ASIL uses an EBT system for tapping molten steel, which is now considered the preferred tapping method in modern EAF steelmaking. A tapping channel is located eccentrically at the furnace bottom. Approximately **50 kg** of EBT Mass is used to pack and seal the tapping hole before charging. The EBT mass acts as a physical barrier that holds the molten steel and slag inside the furnace, ensuring a safe, reliable, and smooth opening of the tap hole without needing external oxygen lancing. When the tapping flap is mechanically opened, the EBT mass drops out due to gravity and the static pressure of the liquid steel bath, allowing molten steel to flow smoothly into the ladle.

The composition of EBT Mass is as follows:

Component	Value
LOI (Loss on Ignition)	5.00%
SiO_2	33.20%
MgO	46.36%

Compared with conventional side tapping, EBT provides the following advantages:

- Lower slag carryover into the ladle
- Better alloy recovery
- Cleaner steel and improved ladle metallurgy
- Reduced phosphorus reversion

7. Raw Materials and Additives Used in EAF

7.1 Typical Material Quantities

The major raw materials used in the ASIL EAF and their typical quantities per heat are listed below:

Material	Typical Quantity per Heat
Scrap (1st Charge)	~16 Ton
Scrap (2nd Charge)	~10 Ton
DRI	10–12 Ton
Lime	~2000 kg
Dolomite	~110 kg
Basic Compound	~300 kg
Met Coke	250–300 kg

Figure 1.7: *Charging bucket*

7.2 Role of Specific Additives

Lime: The most important flux. Functions include slag formation, dephosphorization, basicity control, and protection of refractories.

Dolomite: Supplies MgO, improves slag chemistry, and protects MgO-C refractories.

Basic Compound: Added to improve slag properties and assist refining reactions. Two types are used at ASIL:

- **Basic Compound 2300:** SiO_2: 6.40%, C: 3.9%, CaO: 6.72%, MgO: 77.0%
- **Basic Compound 23F:** SiO_2: 2.60%, C: 0.59%, CaO: 39.20%, MgO: 52.42%

Met Coke: Serves as a carbon source. It facilitates carbon injection, foamy slag generation, and the reduction of FeO ($FeO + C \rightarrow Fe + CO$). The generated CO gas is essential for foamy slag formation.

Oxygen: Injected at approximately **8 kg/cm²** pressure. It accelerates melting, oxidises impurities, promotes decarburization, and generates chemical energy, significantly reducing electrical energy consumption and improving furnace productivity.

8. Complete Electric Arc Furnace Operation

Figure 1.8: *EAF Workstation*

The EAF operation at ASIL converts solid metallic charge materials into molten steel through a precise sequence of melting, refining, and alloying operations. The step-by-step operating procedure for a typical heat is detailed below.

8.1 Furnace Inspection and First Charging

Before charging begins, the furnace is thoroughly inspected to prevent refractory failure, water leakage, arc instability, or slag leakage. Any abnormal wear in the hearth, roof, or sidewalls is repaired using gunning mass before the next heat begins. Following inspection, the roof is swung open and the first scrap bucket containing approximately 16 tonnes of scrap is charged into the furnace. The roof is then closed, the electrodes are lowered, and electrical power is turned on. The initial scrap is melted down to create a liquid bath, which consumes the majority of the electrical energy.

8.2 Second Charging, DRI Addition, and Initial Slag Removal

Once the first charge is melted, the roof is opened again to add a second bucket of scrap (approximately 10-12 tonnes). After that, Direct Reduced Iron (DRI) is fed into the furnace via a conveyor belt system. After the second charge and DRI are completely melted, coking is performed. The initial slag generated at this stage is generally siliceous (high in SiO_2) and is removed from the furnace through the slag door to prepare the bath for proper refining.

8.3 Sequential Refining, Lancing, and Sampling

The refining stage follows a strict sequence of adjustments, sampling, and temperature monitoring. Two specialised sampling techniques are employed:

- **Spectrometer Sample:** liquid steel is poured into a standard frustum-shaped mould after introducing a 4 mm diameter aluminium wire to deoxidise the steel, yielding a dense, high-quality sample for accurate laboratory analysis
- **Brittleness Check:** a secondary sample is cast into a flat plate mould; if the plate fractures easily upon breaking, it indicates high residual carbon and signals the operators that additional oxygen lancing is required

The typical sampling and lancing sequence at ASIL proceeds through 4 to 5 samples as follows:

- **First Sample (S1) = Coking and Slag-Off:** Taken after the first charge is fully melted. Met coke is injected into the bath to generate foamy slag, and the initial siliceous slag formed during melting is removed through the slag door. S1 is taken to establish the baseline bath chemistry after this first slag removal.

- **Second Sample (S2) = Lime + O_2 + Slag-Off:** Based on S1 results, oxygen lancing is initiated along with lime additions to begin decarburization and dephosphorization. The oxidising slag rich in FeO and P_2O_5 is removed through slag-off. S2 is taken to monitor the progress of carbon and phosphorus removal.

- **Third Sample (S3) = Lime + O_2 + Slag-Off:** A second round of oxygen lancing and lime addition is carried out, followed by another slag-off to remove the phosphorus-rich slag and prevent reversion. S3 is taken to confirm that the bath chemistry is approaching the final target parameters.

- **Fourth Sample (S4) = Coking and Slag-Off:** Met coke is injected again to regenerate foamy slag and adjust the carbon level if required, followed by a final slag-off. S4 is taken as the final confirmation of bath chemistry before tapping. Once S4 results are satisfactory and the tapping temperature is confirmed, the furnace is tapped through the EBT system.
- **Fifth Sample (S5):** If the target chemistry is not fully achieved after S4, an additional fifth sample may be taken. This is not a routine step and is only carried out when the results of S4 indicate a deviation that needs to be corrected before tapping.8.4 Foamy Slag Formation, Final Slag Removal, and Tapping.

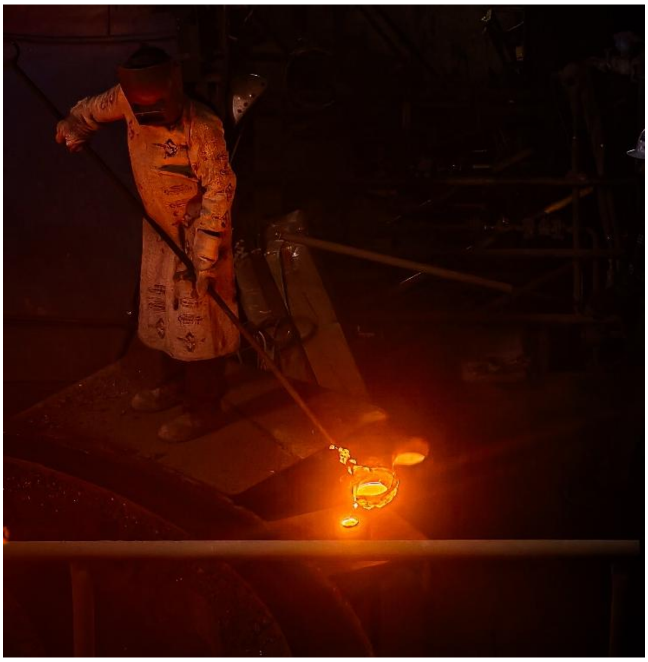

Figure 1.9: *Sample collection*

8.4 Foamy Slag formation, Final slag removal and Tapping

Carbon is injected into the slag to react with FeO **($FeO + C \rightarrow Fe + CO$)**, trapping CO gas and creating a thick, foamy slag that shields the arc, reduces noise, and protects refractories. Once the desired refining especially dephosphorization is achieved, this phosphorus-rich foamy slag is carefully removed to prevent phosphorus from reverting back into the steel.

A final steel sample is taken and the exact tapping temperature is confirmed. The furnace tilts toward the EBT side, and as the EBT flap open

s, the EBT mass drops out, allowing molten steel to flow cleanly into the ladle. During tapping, approximately 50–60% of the required ferroalloys and other metal additions are added directly into the ladle stream before it is sent to the Ladle Refining Furnace.


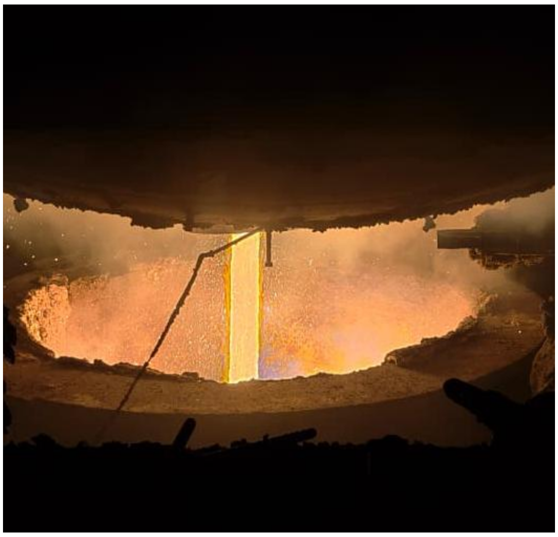

Figure 1.10: *Slag off and Tapping of liquid metal into ladle*

9. EAF Slag Metallurgy

9.1 Importance of Slag Basicity and Oxidising Atmosphere

Slag plays a vital role in steelmaking, performing functions such as dephosphorization, foamy slag generation, arc shielding, refractory protection, and inclusion absorption. For effective refining particularly the removal of phosphorus controlling the slag chemistry and furnace atmosphere is strictly required. A high basicity (high CaO content) and a highly oxidising atmosphere (high FeO content) are mandatory for dephosphorization. The oxidising environment converts phosphorus into P_2O_5, while the high basicity provides the CaO needed to stabilise it into a secure phosphate compound ($3CaO \cdot P_2O_5$) that stays trapped in the slag.

Chemical Reactions in EAF:

- **Silicon Oxidation:** $Si + O_2 \rightarrow SiO_2$
- **Manganese Oxidation:** $Mn + ½O_2 \rightarrow MnO$
- **Iron Oxidation:** $Fe + ½O_2 \rightarrow FeO$
- **Dephosphorization:** $2P + 5O_2 + 3CaO \rightarrow 3CaO \cdot P_2O_5$
- **Foamy Slag Creation:** $C + FeO \rightarrow Fe + CO\uparrow$

If the basicity is too low or the atmosphere loses its oxidising nature, the chemical reaction reverses and phosphorus reverts back into the liquid steel a phenomenon known as phosphorus reversion, resulting in off-spec, brittle steel that fails quality requirements.

9.2 Typical EAF Slag Composition

The typical EAF slag composition is shown below:

Component	Typical Range (%)
CaO	35–50
SiO_2	10–20
FeO	10–30
MgO	5–12
MnO	2–10
Al_2O_3	2–8
P_2O_5	0.5–3

10. Operational Problems and Troubleshooting

The EAF process can encounter several operational challenges. The most common problems, along with their causes and implications, are described below.

10.1 Electrode Breakage

Electrode breakage occurs when heavy un-melted scrap suddenly collapses against the electrodes during the heat, exerting immense mechanical stress. It can also be triggered by severe arc instability during the initial melting phase, leading to erratic electrode movements. Frequent breakages cause significant production delays and increased operational costs due to electrode replacement.

10.2 Excessive FeO Generation

Excessive FeO is primarily caused by prolonged or aggressive over-oxidation of the molten bath during oxygen lancing. This converts valuable metallic iron into iron oxide, directly reducing metallic yield. High FeO levels also make the slag highly aggressive, accelerating chemical erosion of basic refractories. Precise control of the oxygen-to-carbon ratio is strictly required to minimise this costly issue.

10.3 Refractory Wear

Refractory wear is a critical issue driven by the following causes:

- Chemical attacks when slag basicity is improperly balanced or overly oxidised
- Intense direct arc radiation when foamy slag fails to fully cover the arc, severely overheating exposed sidewall panels
- Thermal shocks due to rapid temperature drops when cold scrap is initially charged into the hot furnace

Regular preventive gunning maintenance and meticulous foamy slag practices are essential to mitigate this extensive wear.

10.4 Foamy Slag Collapse

A foamy slag collapse happens when the delicate balance of carbon and oxygen injection is disrupted usually due to low carbon injection rates, causing a loss of CO gas bubbles and rapid foam dissipation. Incorrect slag chemistry such as poor basicity or low temperatures also lowers the slag's viscosity, preventing it from trapping gases. This sudden collapse exposes the electric arc, leading to higher noise levels, extreme heat loss, and immediate refractory damage.

10.5 Slopping

Slopping is characterised by the violent boiling and overflowing of slag from the furnace door, creating significant safety hazards and equipment damage. It is typically caused by a sudden, excessive evolution of CO gas when large amounts of carbon react rapidly with a highly oxidised bath. Operators must carefully monitor the rate of carbon injection and the bath temperature to prevent this explosive gas build-up.

11. Conclusion

The Electric Arc Furnace is the heart of steel production at ASIL. Through the combined application of electrical energy, oxygen injection, carbon injection, slag engineering, and process control, solid metallic charge materials are converted into molten steel suitable for secondary refining. Proper scrap selection, efficient foamy slag practice, controlled oxygen lancing, effective dephosphorization, and clean EBT tapping are essential for achieving high productivity and superior steel quality. The EAF therefore serves as a highly integrated metallurgical reactor that links scrap recycling with modern steel production and forms the foundation of the entire steelmaking route at ASIL.

CHAPTER 2

LADLE REFINING FURNACE (LRF)

&

VACUUM DEGASSING

(VD)

1. Introduction to Secondary Refining

Steel produced in the Electric Arc Furnace undergoes primary refining where the major objectives are melting of scrap, decarburization, dephosphorization, and achievement of the desired bath chemistry. However, the molten steel obtained after EAF refining is not sufficiently clean for direct casting. The steel still contains dissolved gases (primarily hydrogen and nitrogen), non-metallic inclusions, temperature variations, and minor deviations from the target chemical composition.

To achieve the stringent quality requirements of engineering steels, alloy steels, and carburising grades, secondary refining operations are carried out. At Aarti Steel International Limited, secondary refining is performed through the Ladle Refining Furnace (LRF) followed by Vacuum Degassing (VD). The complete secondary refining route followed at the plant is:

EAF	LRF	VD	CCM

The main objectives of secondary refining are:

- Precise chemistry control of all alloying elements
- Temperature adjustment and superheating to required casting temperature
- Deoxidation of molten steel to reduce dissolved oxygen
- Desulphurisation to achieve low sulphur specifications
- Inclusion modification and removal for cleaner steel
- Removal of dissolved hydrogen and nitrogen gases
- Improvement of overall steel cleanliness and uniformity
- Preparation of steel for continuous casting at the CCM

2. Transfer of Molten Steel EAF Tapping

After completion of EAF refining, the molten steel is tapped through the Eccentric Bottom Tapping (EBT) system into a preheated ladle. The EBT system allows clean tapping from the bottom of the furnace, minimising slag carry-over into the ladle.

Ladle preheating is an absolutely essential operation because direct contact between molten steel at approximately 1640°C and a cold ladle would result in excessive and uncontrollable temperature loss. Ladles are therefore preheated to approximately 900–1100°C before receiving molten steel. The benefits of ladle preheating include:

- Reduced thermal shock to the ladle refractory lining
- Prevention of excessive temperature drop during tapping
- Minimisation of moisture that could increase hydrogen pickup
- Longer refractory life

The following standard operating parameters are maintained during EAF tapping:

Parameter	Value / Range
Heat Size	33 ± 3 MT
Tapping Temperature	1640 ± 30°C
Dissolved Oxygen at Tapping	~400 ppm
Argon Purging Pressure	2–5 kg/cm²
Tapping System	Eccentric Bottom Tapping (EBT)

Simultaneously with the start of tapping, argon purging is initiated through the ladle bottom porous plugs. Argon, being an inert gas, does not react with molten steel. Its continuous bubbling performs several important metallurgical functions:

- Bath homogenisation uniform temperature and composition throughout the liquid steel
- Removal of non-metallic inclusions by flotation to the slag surface
- Prevention of local overheating or cold zones
- Enhanced slag-metal reactions for improved refining

3. Ferroalloy and Flux Additions During Tapping

The initial chemistry correction begins during tapping itself. Approximately 50–60% of the total required ferroalloys are added into the ladle during the tapping stream. The turbulence generated by the steel stream entering the ladle provides excellent mixing conditions, resulting in:

- Higher alloy recovery due to vigorous mixing
- Better dissolution of alloying elements
- More uniform distribution throughout the melt
- Reduced treatment time at the LRF

The ferroalloys used are listed in the table below:

Ferroalloy	Symbol	Metallurgical Purpose
Ferro Manganese	FeMn (HC/MC)	Increases Mn content; improves hardenability and strength
Silico Manganese	SiMn	Combined Mn and Si source; acts as deoxidizer
Ferro Silicon	FeSi	Silicon addition and primary deoxidation
Ferro Chromium	FeCr (HC/LC)	Increases Cr; improves wear resistance and hardenability
Ferro Molybdenum	FeMo	Improves strength, creep resistance, and hardenability
Nickel	Ni	Enhances toughness and impact resistance
Ferro Titanium	FeTi (70%)	Grain refinement and nitrogen fixation

Along with ferroalloys, the following fluxes are added during tapping to begin slag conditioning:

- **Lime (CaO):** 100–300 kg primary basicity agent for sulphur removal
- **Basic Compound:** 50–100 kg slag conditioning and foaming
- **Coke:** as per grade requirement carbon source
- **Aluminium (Al-bar):** 0.5–2 kg/MT strong deoxidiser to reduce dissolved oxygen

4. Ladle Refining Furnace (LRF) Detailed Operation

After tapping, the ladle is transferred to the Ladle Refining Furnace station. Unlike the EAF, whose primary function is melting, the LRF focuses exclusively on achieving the exact chemistry and temperature required for casting. The LRF is equipped with the following:

- Three graphite electrodes (10 feet each) arranged in a delta configuration for arc heating
- A water-cooled roof that lowers over the ladle to contain the atmosphere
- Argon purging system through bottom porous plugs of the ladle
- Wire feeder for injecting aluminium, CaSi, and sulphur wires
- Immersion thermocouple for temperature measurement
- Sampling system for spectrometric analysis

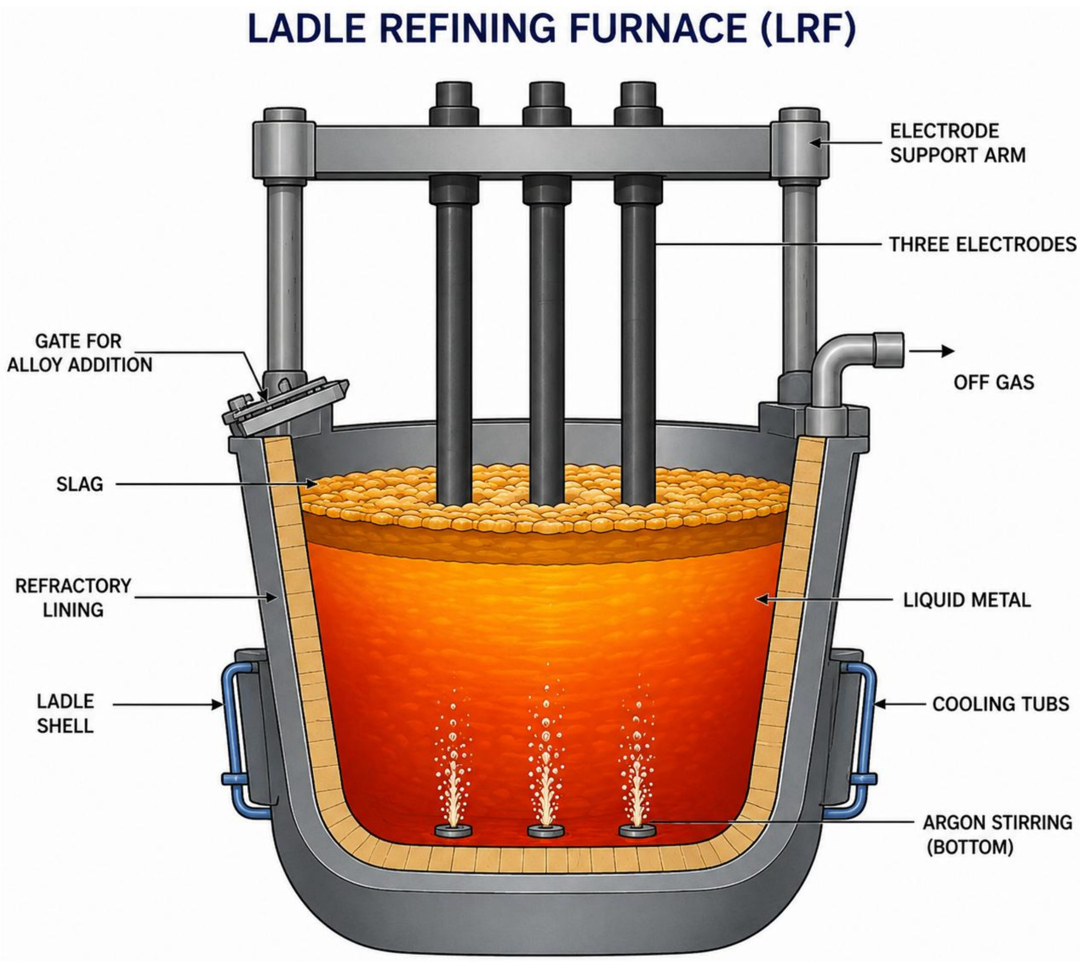

Figure 2.1: *Ladle Refining Furnace Schematic Diagram*

Figure 2.2: *Ladle Refining Furnace*

4.1 Step-by-Step LRF Processing

Step 1 Positioning and Setup

The ladle is placed under the LRF roof using an overhead crane. The roof is then lowered and graphite electrodes are aligned precisely over the bath. Argon purging through the bottom plugs is adjusted to a pressure of 2–5 kg/cm² and a flow rate of 20–40 LPM to ensure gentle but effective stirring and homogeneous temperature distribution.

Step 2 Arc Heating

Power is switched on and the graphite electrodes generate an electric arc that heats the molten steel bath, compensating for temperature losses during tapping and transportation. The arc heating is controlled through the following tap settings:

- **Tap 1:** 210 V
- **Tap 2:** 185 V

After approximately 8–10 minutes of arcing, the first temperature measurement is taken and the first steel sample (R1) is collected.

Step 3 First Sample (R1) and Chemistry Analysis

The R1 sample is sent immediately to the laboratory for spectrometric analysis, determining the full composition of all critical elements: Carbon, Manganese, Silicon, Phosphorus, Sulphur, Chromium, Molybdenum, Nickel, Copper, and Aluminium.

Step 4 Chemistry Correction

The obtained chemistry is compared with the target composition for the steel grade being produced and additional ferroalloys are calculated and added accordingly. The sampling and correction sequence follows an iterative approach:

- **R1:** Initial analysis → Major alloy additions
- **R2:** Verify → Correction additions
- **R3:** Fine-tune → Minor adjustments
- **R4:** Final verification before VD

Multiple additions are made in steps rather than all at once for the following reasons:

- Better alloy recovery by avoiding over-addition
- Reduced oxidation losses during each addition
- Easier chemistry control prevents overshooting the target
- Compensation for slag pickup and oxidation losses between additions

5. Argon Stirring and Slag Refining

At the LRF station, argon gas is continuously injected through porous plugs located at the bottom of the ladle throughout the entire treatment period. The bubbling action performs the following critical metallurgical functions:

- Uniform distribution of alloying elements throughout the bath
- Homogeneous temperature eliminates hot or cold spots
- Enhanced slag-metal reactions at the interface
- Flotation of non-metallic inclusions toward the slag layer
- Continuous exposure of fresh steel to the slag for desulphurisation
- Improved overall cleanliness of steel before casting

Without proper argon stirring, temperature gradients and composition variations would develop inside the molten steel bath, resulting in inconsistent billet quality during continuous casting.

A basic refining slag is maintained over the molten steel surface throughout the entire LRF treatment. The slag absorbs non-metallic inclusions such as Al_2O_3, SiO_2, and sulphides, which float upward through argon stirring and are retained within the slag layer. It acts as a physical barrier preventing atmospheric oxygen from re-entering the molten steel and promotes sulphur removal through the reaction [S] + (CaO) → (CaS) + [O]. The slag layer also acts as a thermal blanket, significantly reducing heat loss by radiation from the molten steel surface. Lime (CaO) is one of the most important constituents of the refining slag, responsible for maintaining the high basicity (CaO/SiO_2 ratio of 3.0–4.0 or higher) required for effective sulphur removal.

6. Temperature Control at LRF

Temperature control is equally as important as chemistry control in the LRF. Heat losses occur continuously throughout treatment due to radiation from the steel surface, heat absorbed by ferroalloy additions (which arrive at room temperature), cooling during sampling, conduction through ladle walls and bottom, and transportation time between stations. The temperature targets maintained are:

Condition	Target Temperature
Standard LRF Target	1550 ± 30°C
Lifting Temperature (Non-VD Heats)	Liquidus + 50 ± 20°C
Lifting Temperature (VD Heats)	Liquidus + 130 ± 20°C

For VD-treated heats, a significantly higher superheat (130°C above liquidus rather than 50°C) is deliberately maintained at LRF completion because substantial temperature losses occur during the vacuum treatment process typically 30–50°C and this must be accounted for to ensure the steel arrives at the CCM at the correct casting temperature. Temperature measurements are periodically performed using immersion thermocouples dipped directly into the molten steel bath, with arc heating applied as needed to restore or maintain the target.

7. Wire Feeding Operations

After achieving the desired chemistry confirmed by the R3 or R4 sample, wire feeding operations are carried out as the final step of LRF treatment. Wire feeding allows precise micro-additions of reactive elements that would be difficult to control through bulk ferroalloy additions.

Aluminium is the most powerful deoxidiser used in steelmaking. Despite earlier bulk aluminium additions during tapping, residual dissolved oxygen remains in the steel. Aluminium wire is injected through a guide tube directly into the molten steel at a controlled rate, driving the reaction $2Al + 3[O] \rightarrow Al_2O_3$. This reduces dissolved oxygen content from typically 100–200 ppm down to the target level of approximately 20 ppm. The Al_2O_3 particles formed float upward and are absorbed by the refining slag.

CaSi wire feeding is one of the most important final treatments in LRF processing. Pure calcium is injected into the steel via CaSi wire, and calcium chemically modifies the hard, solid alumina inclusions through the reaction:

$$\mathbf{Ca + Al_2O_3 \rightarrow \text{Calcium Aluminate (liquid at steelmaking temperature)}}$$

The key benefits of CaSi wire treatment include:

- Solid alumina inclusions (hard, angular) are transformed into liquid, globular calcium aluminates
- Globular inclusions are less damaging to mechanical properties and less prone to nozzle clogging
- Improved castability reduced risk of submerged entry nozzle (SEN) blockage at CCM
- Better steel cleanliness for demanding engineering applications
- Sulphide shape control calcium also modifies MnS inclusions to globular CaS form

For certain special grades that require a specified minimum sulphur level, such as free-machining steels, sulphur wire is injected to achieve the exact target sulphur content. This is only done for specific grade requirements and not as a routine operation.

8. Ferroalloy Addition Calculation

Accurate ferroalloy addition calculation is critical for achieving target chemistry while minimising cost and wastage. The quantity of ferroalloy required is determined by the heat size, current composition from spectrometric analysis, target composition from the grade specification, alloy recovery factor, and ferroalloy composition. The general formula used is:

$$\mathbf{Ferroalloy\ Required\ (kg)} = \frac{(\mathbf{\Delta\%} \times \mathbf{Heat\ Weight} \times \mathbf{1000})}{(\mathbf{Recovery} \times \mathbf{Alloy\ Content})}$$

Where;

Δ% is the required increase in element percentage (Target minus Current)

Heat Weight is the total liquid steel weight in tonnes

Recovery is the fractional alloy recovery (for example 0.85 for 85%)

Alloy Content is the fraction of the desired element in the ferroalloy (for example 0.65 for 65% Cr in FeCr).

As an example, for Ferro Chromium addition: with a heat weight of 33 tonnes, current Cr of 0.90%, target Cr of 1.20%, FeCr content of 65%, and recovery of 85%, the FeCr required calculates to approximately 180 kg. Similarly for Ferro Manganese: with a heat weight of 33 tonnes, current Mn of 1.00%, target Mn of 1.30%, FeMn content of 75%, and recovery of 90%, the FeMn required calculates to approximately 150 kg.

For rapid shop-floor calculations, a standard quick-reference chart is used to estimate alloy additions required for increments of 0.01% in any element:

Ferroalloy	Addition per 0.01% Increase (kg)
FeMn (High Carbon)	5 kg
FeMn (Medium Carbon)	5 kg
FeCr (High Carbon)	6 kg
FeCr (Low Carbon)	6 kg
FeSi	7 kg
FeMo	6 kg
Nickel	3 kg
FeTi (70%)	7 kg
S wire (13 mm dia)	4 mtr per 0.001%
Al wire (9 mm dia)	9 mtr per 0.001%

9. Vacuum Degassing (VD)

Vacuum Degassing (VD) is a secondary refining process used to improve steel cleanliness by removing dissolved gases and further reducing non-metallic inclusions. This process is essential for producing high-quality engineering steels that require excellent internal cleanliness, low hydrogen content to prevent hydrogen-induced cracking, and low nitrogen to maintain ductility and toughness.

9.1 Transfer from LRF to VD

Once the desired chemistry and temperature are achieved at the LRF (with sufficient superheat of Liquidus + 130°C for VD heats), the argon purging pipe is disconnected from the bottom plug, the overhead crane lifts and transports the ladle to the VD station, and transfer must be completed quickly to minimise temperature loss.

9.2 Principle of Vacuum Degassing

Vacuum Degassing is based on the fundamental physical principle known as Sievert's Law, which states that the solubility of a gas in liquid metal is directly proportional to the square root of the partial pressure of that gas above the melt: $[\text{Gas\%}] \propto \sqrt{P(\text{gas})}$. When molten steel is exposed to a high vacuum environment (very low partial pressure of H_2 and N_2), dissolved gases become thermodynamically unstable and escape from the steel into the gas phase above. The very low pressures achieved (≤ 1 mbar) drive virtually all dissolved hydrogen and a significant fraction of nitrogen out of the steel.

9.3 Gases Removed During Vacuum Treatment

Hydrogen is removed through the reaction $[H] \rightarrow \frac{1}{2}H_2\uparrow$, which prevents hydrogen flakes (internal cracking known as 'white spots' or 'fish eyes') in rolled products particularly critical for thick sections and bearing steel grades. Nitrogen is removed through $[N] \rightarrow \frac{1}{2}N_2\uparrow$, as excess nitrogen reduces ductility and impact toughness, and nitrogen combines with aluminium or titanium to form nitrides that pin grain boundaries and cause brittleness. Residual dissolved oxygen is further reduced under vacuum conditions, as vacuum treatment reduces oxygen activity and improves overall steel cleanliness.

9.4 Vacuum Generation and Equipment

The VD process requires specialised equipment to achieve and maintain the very low pressures required for effective degassing. Steam ejector pumps arranged in series stages reduce pressure progressively; the ladle is placed inside a sealed VD chamber; the chamber lid is sealed airtight before vacuum pumps are activated; and pressure is reduced from atmospheric (approximately 1013 mbar) to the target operating vacuum. The plant operating parameters are:

VD Parameter	Plant Value
Target Vacuum Level	< 1 mbar
Observed Operating Vacuum	~1 mbar
Time to Achieve Vacuum	Within 10 minutes
Hold Time Under Vacuum	As per grade specification

9.5 Argon Stirring During VD

Argon stirring through the bottom porous plugs is continued throughout the vacuum treatment at a carefully controlled rate. During vacuum, argon stirring continuously exposes fresh steel surface to the vacuum atmosphere, dramatically accelerating gas removal. The argon bubbles expand massively under vacuum (volume increases by approximately 1000 times), providing very vigorous mixing that causes inclusions to collide, agglomerate, and float more quickly to the slag surface. Stirring also maintains bath temperature homogeneity and promotes effective slag-metal contact for any remaining desulphurisation. The argon flow rate during VD is carefully controlled to avoid excessive splashing that could cause refractory erosion.

9.6 Inclusion Removal During VD

Vacuum treatment is particularly effective at inclusion removal through a synergistic mechanism:

- The CO reaction: dissolved carbon reacts with dissolved oxygen at very low partial pressures, generating CO bubbles that assist in stirring and inclusion flotation
- Expanding argon bubbles attach to inclusions and carry them upward to the slag layer
- The slag layer at the top absorbs and traps floated inclusions

The result is significantly cleaner steel with fewer and smaller non-metallic inclusions.

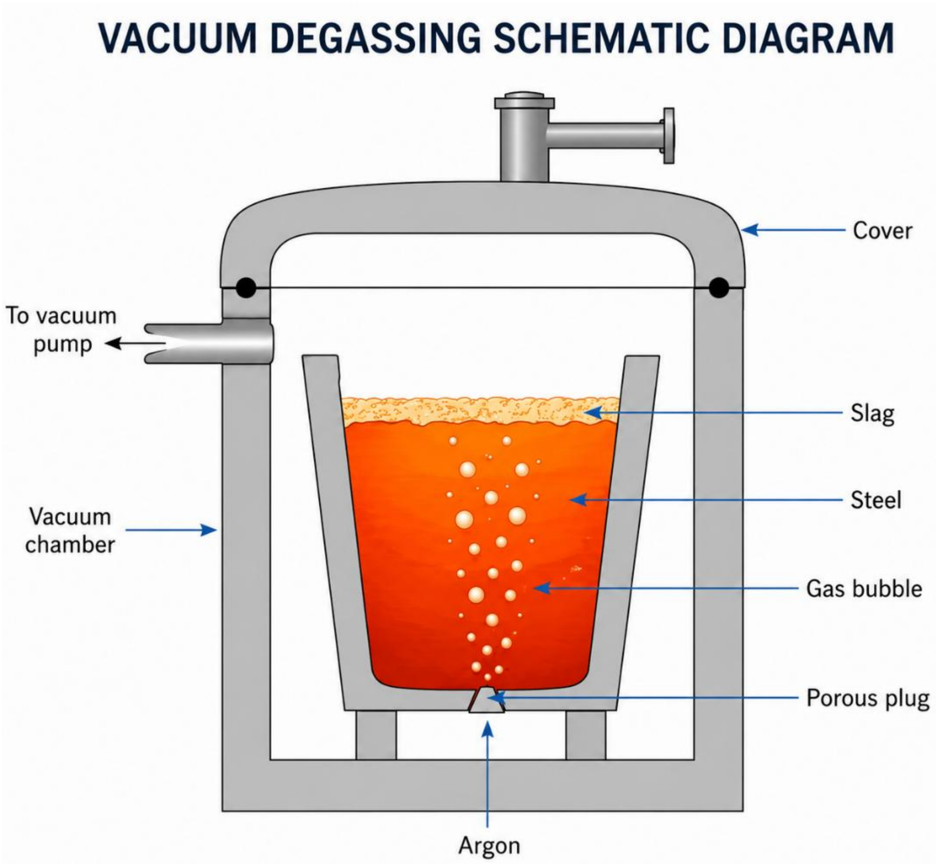

Figure 2.3: *Vacuum degassing Schematic Diagram*

Figure 2.4: *Vacuum degassing Chamber*

9.7 Importance of Redex-702

After the completion of vacuum degassing and breaking of the vacuum, Redex-702 is applied over the molten steel surface before the ladle is transferred to the CCM. It is a refractory-grade covering compound with a high Al_2O_3 content of 70.12%, which gives it a very high melting point (~2072°C), ensuring it remains stable and does not contaminate the steel below. Its primary purpose is to prevent reoxidation of the clean degassed steel surface and minimise temperature loss during transfer to the CCM. This is particularly critical after VD treatment since the steel has been refined to very low oxygen levels and any atmospheric contact would immediately reverse this achievement. Its composition is:

Constituent	Value
Moisture	1.20%
Carbon (C)	2.19%
SiO_2	20.40%
Al_2O_3	70.12%

The low moisture content of 1.20% is also important any significant moisture would decompose at steelmaking temperatures and reintroduce hydrogen back into the steel, directly undoing the degassing work performed during VD treatment.

Figure 2.6: *Redex layer on liquid metal*

10. Final Sampling, Quality Verification, and Transfer to CCM

After completion of VD treatment and breaking of the vacuum, a final steel sample is collected and sent to the laboratory for complete spectrometric analysis of all elements. Temperature is verified using an immersion thermocouple, all elements are checked against the grade specification, and the typical target after VD for dissolved oxygen is ≤35 ppm.

After successful completion of VD treatment and final quality approval by the metallurgist, the ladle is transferred to the Continuous Casting Machine. The complete casting sequence proceeds through five stages: the ladle holds refined molten steel; the tundish acts as a buffer vessel providing additional inclusion separation and flow rate control; the copper mould is where initial solidification begins; the secondary cooling zone completes solidification; and the final billets are cut and transferred for inspection and rolling.

11. Conclusion

The LRF and VD sections play an absolutely vital role in producing clean, high-quality steel suitable for demanding engineering applications. The secondary refining process transforms partially refined EAF steel into a precisely controlled, inclusion-free, gas-free steel ready for continuous casting.

Summary of LRF Achievements

- Precise chemistry control through iterative sampling (R1 → R4) and calculated ferroalloy additions
- Temperature adjustment to exact casting requirements using graphite electrode arc heating
- Effective desulphurisation through high-basicity CaO-rich slag maintained under argon stirring
- Final deoxidation through aluminium wire feeding, reducing dissolved oxygen to target levels
- Inclusion modification through CaSi wire feeding, converting harmful hard alumina to benign globular calcium aluminates

Summary of VD Achievements

- Removal of dissolved hydrogen to prevent hydrogen flaking in rolled products
- Partial removal of dissolved nitrogen to maintain steel ductility and toughness
- Further reduction in dissolved oxygen content under vacuum conditions
- Significant removal of non-metallic inclusions through argon stirring and inclusion flotation
- Achievement of oxygen level ≤35 ppm for clean steel production

Through the combined and coordinated operations of tapping, LRF treatment, and VD processing, the molten steel achieves all the required quality parameters for engineering steel production. The success of this secondary refining process is the foundation of consistent billet quality and is essential to ASIL's ability to produce high-value carburising and engineering steel grades.

CHAPTER 3

CONTINUOUS CASTING MACHINE

(CCM)

1. Introduction

The Continuous Casting Machine (CCM) represents the final and most critical stage of steelmaking, where refined molten steel is transformed into solid billets of a defined cross-section. The process demands precise control over temperature, flow, cooling intensity, and mould level to ensure that the resulting billets possess sound internal structure and acceptable surface quality. Any deviation in process parameters during casting can lead to defects that are difficult or impossible to correct downstream.

Molten steel produced in the Steel Melting Shop is transferred into a ladle, from which it is routed through a ladle shroud into the tundish. The tundish acts as an intermediate reservoir, distributing molten steel uniformly through the Submerged Entry Nozzle (SEN) into the water-cooled copper mould below. Inside the mould, the strand begins to solidify at its outer surface. It then progresses downward through the secondary cooling zone, is guided by withdrawal rollers, and finally reaches the torch cutting station where billets of the required length are produced.

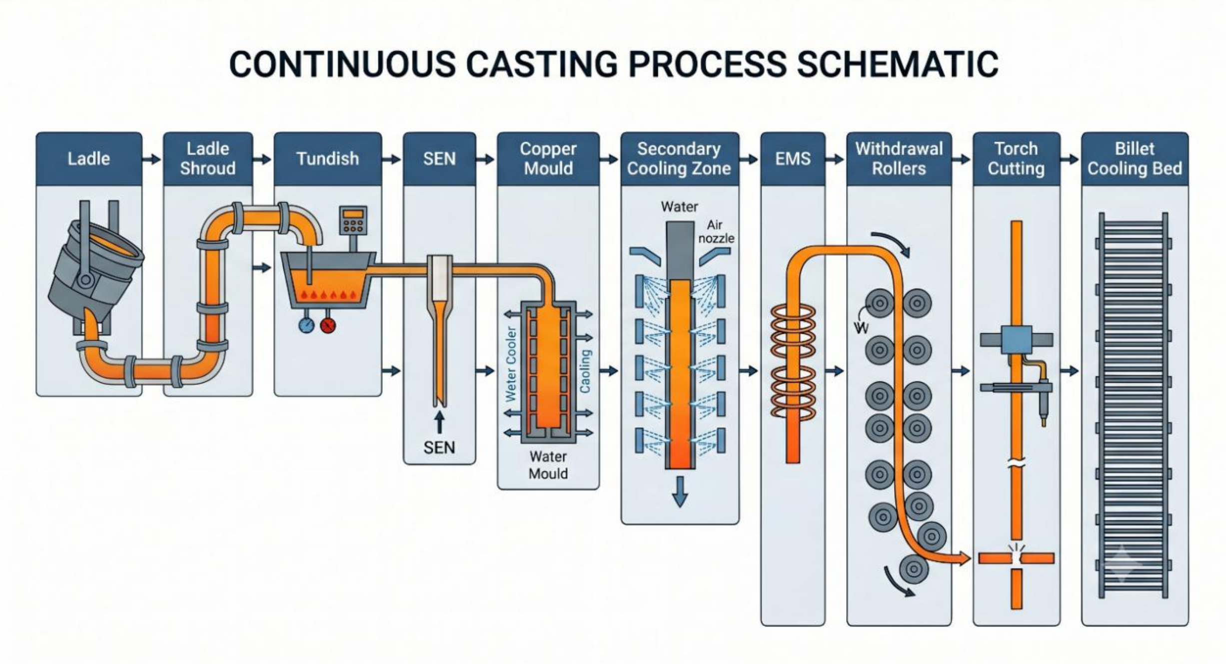

Figure 3.1: *CCM Process layout*

2. Tundish

The tundish is a refractory-lined intermediate vessel that bridges the gap between the ladle above and the casting mould below. Beyond simply holding molten steel, it performs several important metallurgical functions: it ensures a continuous and steady flow of steel to all casting strands, maintains a constant ferrostatic head, promotes the flotation of non-metallic inclusions towards the slag layer, minimises turbulence, and enables sequence casting by allowing ladle exchanges without interrupting the flow of metal.

2.1 Tundish Geometry

The tundish used for billet casting is trapezoidal in cross-section, a geometry specifically chosen to minimise stagnant dead zones and maximise the residence time of molten steel. A longer residence time allows more inclusions to float to the surface and be absorbed into the tundish flux, improving the cleanliness of the steel entering the mould. the tapered design also promotes smoother metal flow towards the SEN while reducing turbulence at the outlet.

Based on plant measurements, The principal dimensions of the tundish are as follows:

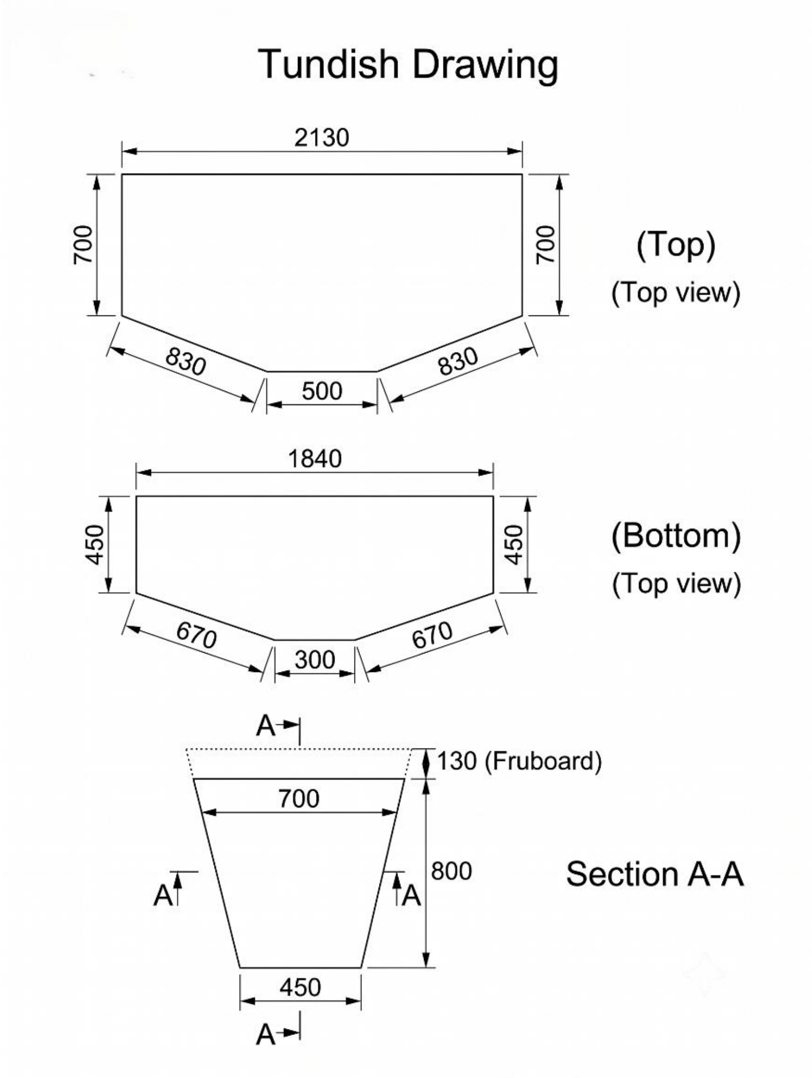

Figure 3.2: *Tundish geometry schematic diagram*

View	Dimension	Value
Top View	Overall Length	2130 mm
	Width	700 mm
	Inclined Bottom Portion	830 mm
	Central Flat Portion	500 mm
Bottom View	Length	1840 mm
	Width	450 mm
	Inclined Portion	670 mm
	Central Flat Portion	500 mm
Side View	Working Depth	800 mm
	Freeboard	120 mm
	Top Width	700 mm
	Bottom Width	450 mm

2.2 Turbopad

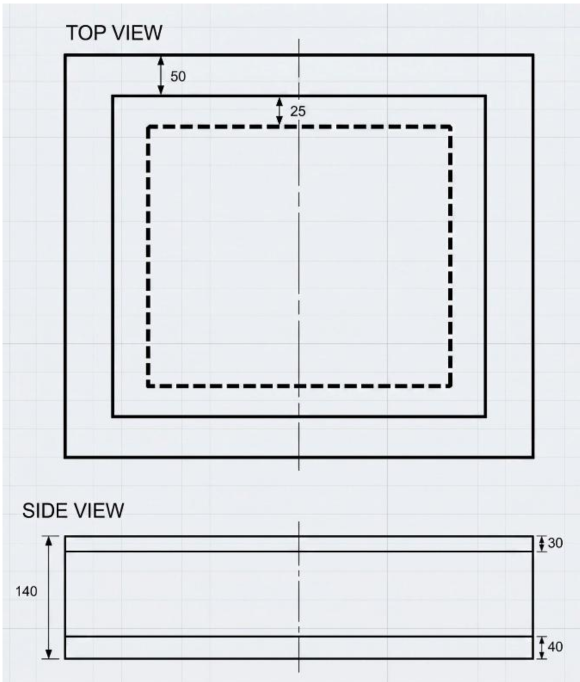

Figure 3.3: *Turbopad*

A Turbopad is installed on the tundish floor directly beneath the ladle shroud, at the point where the incoming stream of molten steel impacts the vessel. When steel exits the ladle at high velocity, it carries considerable kinetic energy. If this jet were to strike the bare tundish floor, it would generate violent turbulence, causing slag entrainment, refractory erosion, and reoxidation of the steel. The Turbopad absorbs this kinetic energy and redirects the flow horizontally, converting a turbulent downward jet into a calmer, more laminar horizontal flow, which greatly improves inclusion flotation and protects the tundish bottom refractory from erosion damage.

The Turbopad is approximately 400 mm long, 350 mm high, with a central cavity of 250 mm and wall thickness of 50 mm. Its chemical composition is dominated by high alumina (Al_2O_3 85.33%), which provides excellent refractoriness and erosion resistance under continuous exposure to molten steel at casting temperatures.

2.3 Refractory Coating (Mud Spraying)

Figure 3.4: *Mud spraying on tundish refractory*

Before every casting sequence, the tundish interior is thoroughly cleaned, residual skull and old slag are removed, and the refractory lining is inspected for surface cracks or damage. A uniform layer of refractory coating commonly referred to as tundish mud is then sprayed over the permanent lining. This coating protects the permanent refractory from direct contact with molten steel, minimises refractory wear, reduces heat loss, and prevents steel penetration into the lining material. After casting, the mud allows easy removal of the solidified steel skull, extending the service life of the tundish.

2.4 Submerged Entry Nozzle (SEN)

After the application of the refractory coating and installation of the turbopad, the Submerged Entry Nozzle (SEN) is fixed at the tundish outlet. The SEN is carefully positioned and secured using a specially formulated refractory sealing mud, which is applied around the seating area to ensure a leak-proof joint and provide mechanical stability during casting. The SEN serves as the passage through which molten steel flows from the tundish into the copper mould below the slag layer. By keeping the steel stream submerged beneath the mould flux, the SEN prevents direct contact of molten steel with the atmosphere, thereby minimising reoxidation, nitrogen pickup, and slag entrainment. During casting, the SEN immersion depth is maintained at approximately 110 ± 10 mm to ensure stable flow conditions inside the mould.

Figure 3.5: *SEN*

2.5 Drying and Preheating

After the SEN has been installed and sealed, the complete tundish assembly undergoes a controlled preheating process using gas burners. This operation dries the refractory coating and the sealing mud surrounding the SEN, allowing them to develop sufficient mechanical strength and ensuring a firm bond between the nozzle and the tundish. Preheating also removes residual moisture the presence of moisture is highly undesirable, as it can rapidly vaporise upon contact with molten steel and generate steam explosions. Adequate preheating also minimises thermal shock during steel pouring, reduces heat loss from the molten steel, and enhances the service life of the tundish lining and nozzle assembly.

2.6 Ladle Shroud

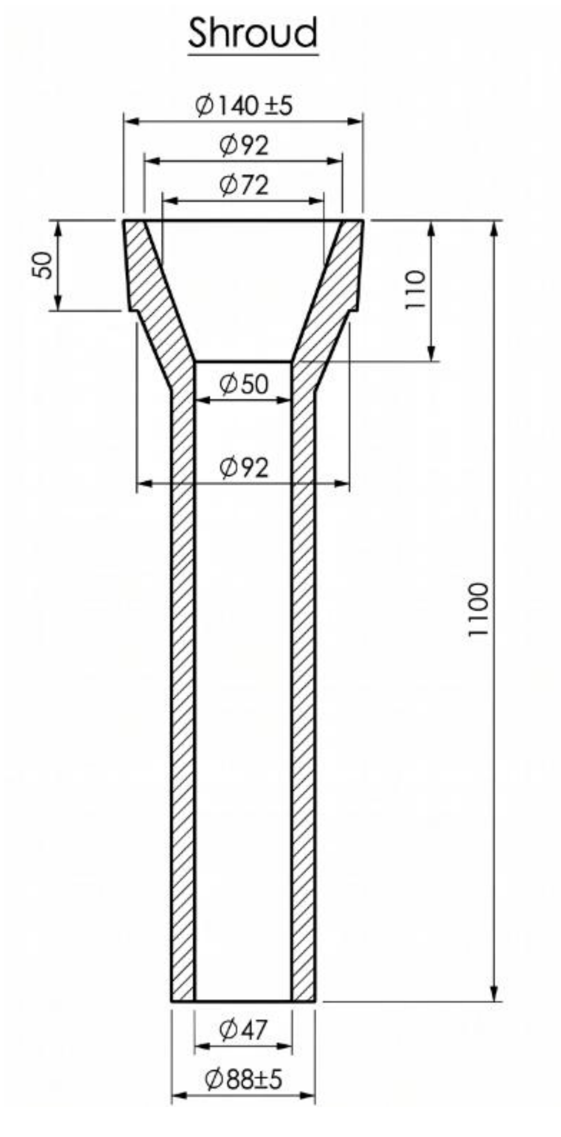

Figure 3.6: *Ladle shroud*

The shroud has an overall length of 1100 mm, with a top flange having a maximum outer diameter of Ø140 ± 5 mm and an initial inner diameter of Ø92 mm. Internally, a divergent nozzle tapers from a top inner diameter of Ø72 mm down to a throat opening of Ø50 mm at a depth of 110 mm. The main straight shank continues with an outer diameter of Ø88 ± 5 mm and an internal bore of Ø47 mm through to the exit. It is constructed from a composite refractory material consisting of 47–53% alumina (Al_2O_3) for high-temperature structural strength, 35–36% carbon for thermal shock resistance, and 9–14% silica (SiO_2) as a stabilising agent. The ladle shroud serves as a critical conduit connecting the ladle nozzle to the tundish, providing a fully sealed and protected pathway for transferring molten steel. By shielding the metal stream from atmospheric oxygen and nitrogen, it effectively prevents reoxidation, nitrogen pickup, and the formation of non-metallic inclusions, while also minimising hazardous splashing to ensure a stable, controlled flow.

3. Casting Operation

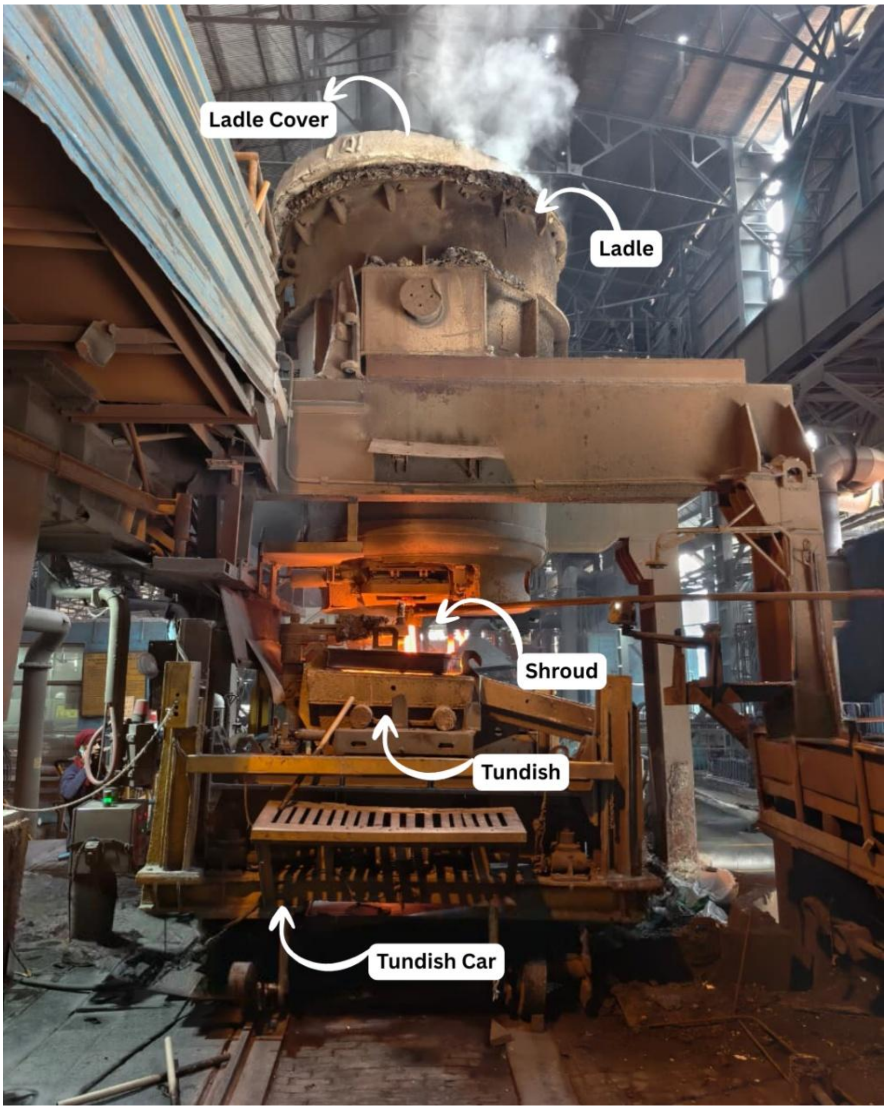

Figure 3.7: *CCM Workstation*

3.1 CCM Preparation before Casting

Before the ladle is positioned and casting begins, a comprehensive check of all machine systems is carried out to ensure operational readiness. Every subsystem must be confirmed as fully functional before any molten steel is received, since a failure during casting can lead to breakouts, strand misalignment, or uncontrolled solidification conditions. The pre-casting checks include:

- Hydraulic system
- Mould oscillation mechanism
- Primary cooling water supply
- Secondary cooling water circuits
- Electromagnetic Stirrer (EMS) current and frequency settings
- Withdrawal roller table
- Dummy bar position verification
- Auto mould level controller calibration
- Argon gas supply lines

3.2 Start of Casting

With the tundish preheated and all CCM systems verified, the ladle containing molten steel is positioned on the ladle turret above the tundish. The slide gate at the ladle bottom is opened to allow steel to flow through the ladle shroud into the tundish. Argon purging is initiated simultaneously to prevent reoxidation. As the steel level in the tundish builds up, the dummy bar seals the bottom of the mould initially, and casting powder is added to the mould surface. Casting is initially operated in manual mode, with the operator monitoring and adjusting the mould level. Once the steel level stabilises at the preset value and the system confirms consistent conditions, control is transferred to the automatic mould level controller for sustained operation.

3.3 Auto Mould Level Control

Maintaining a constant and stable molten steel level inside the copper mould is one of the most critical requirements for producing sound billets. The Auto Mould Level Controller uses a radioactive source and scintillation counter to continuously monitor the steel level in the mould. Fluctuations in level are detected and corrected by adjusting the SEN slide gate opening in real time. A stable mould level ensures uniform shell formation around the mould perimeter, prevents the entrapment of casting powder into the steel, and supports consistent billet surface quality. Any significant level variation can lead to surface depressions, longitudinal cracks, or slag inclusions in the billet.

3.4 Casting Powder

Casting powder is added continuously over the surface of the molten steel inside the mould throughout the casting sequence. At the high temperatures prevailing in the mould, the powder melts to form a liquid slag film that fills the gap between the solidifying billet shell and the copper mould wall. This liquid film acts as a lubricant during mould oscillation, provides thermal insulation to the steel surface, absorbs non-metallic inclusions that rise from the liquid steel, and prevents atmospheric oxidation of the steel surface.

Powder Type	Application
STPS21 DS	Medium carbon steel grades
STPS21 MD	Low carbon steel grades
STP250	120 × 120 mm billet casting

3.5 Mould Oscillation

During continuous casting, the copper mould is not kept stationary but is made to oscillate vertically with a controlled frequency and stroke throughout the casting process. This oscillatory motion is synchronised with the withdrawal of the strand and is achieved through a hydraulic or mechanical oscillation mechanism. The primary purpose of mould oscillation is to prevent the solidifying steel shell from adhering to the inner wall of the copper mould during the initial stages of solidification. Since the shell is relatively thin and weak immediately after formation, direct sticking to the mould surface can lead to shell rupture and severe casting accidents. The oscillating movement creates a periodic relative motion between the mould and the strand, allowing mould flux to penetrate the gap between them and form a lubricating slag film that facilitates smooth withdrawal.

Mould oscillation also plays a crucial role in controlling heat transfer and maintaining stable shell growth within the mould. The liquid mould flux infiltrating the mould-shell interface acts as both a lubricant and a thermal resistance layer, ensuring uniform heat extraction and reducing local thermal stresses. Proper selection of oscillation frequency and stroke minimises friction between the mould and the strand, decreases the probability of longitudinal surface cracks, and significantly reduces the risk of breakout. The periodic motion produces characteristic surface impressions known as oscillation marks, which when oscillation parameters are properly optimised remain shallow and uniform without adversely affecting product quality.

3.6 Electromagnetic Stirrer (EMS)

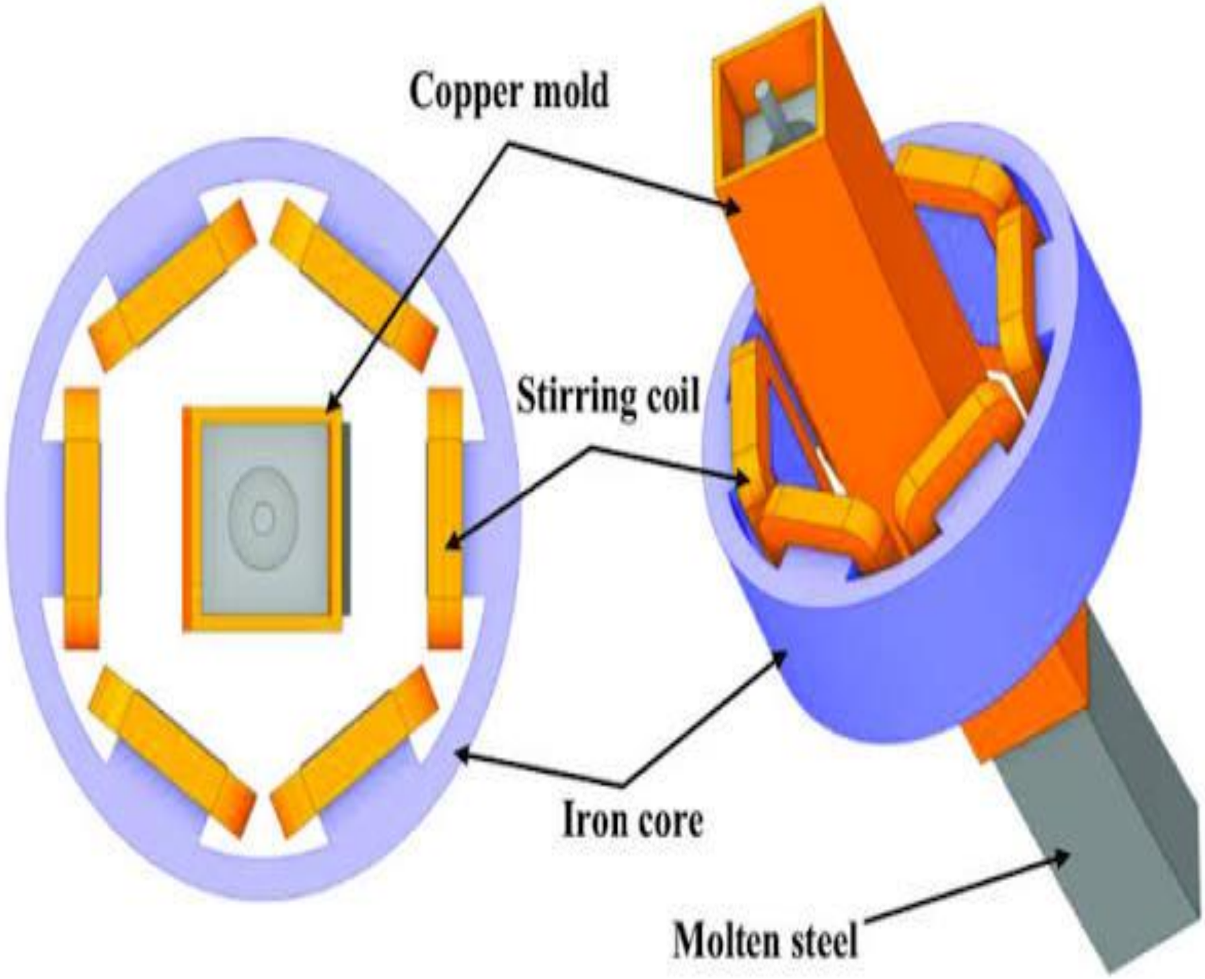

Figure 3.8: *Structure Diagram of Electromagnetic Stirring*

The Continuous Casting Machine is equipped with an Electromagnetic Stirrer (EMS) located in the secondary cooling region. The EMS consists of electromagnetic coils that generate a rotating magnetic field around the strand, inducing controlled motion in the liquid core without any physical contact. The stirring intensity is regulated by controlling the current and frequency supplied to the electromagnets, enabling stable operation under different casting conditions.

Depending on the position along the strand where stirring is applied, EMS is classified into three types, Mould EMS (M-EMS), Strand EMS (S-EMS), and Final EMS (F-EMS), each targeting a different zone of the solidifying strand and serving a distinct metallurgical purpose. The comparison of all three types is presented in the table below:

Feature	M-EMS (Mould EMS)	S-EMS (Strand EMS)	F-EMS (Final EMS)
Location	Inside or around the copper mould	Below the mould in secondary cooling zone	Near the final solidification zone (end of liquid pool)
Target Zone	Fully liquid steel inside the mould	Partially solidified strand with large liquid core	Last remaining solute-enriched liquid at the centreline
Primary Purpose	Improve surface quality and shell uniformity	Reduce columnar grain growth; promote equiaxed structure	Reduce centreline segregation and porosity
Metallurgical Effect	Breaks initial dendrites; promotes fine equiaxed grains near surface; improves temperature homogeneity in mould	Fragments columnar dendrites; enlarges equiaxed zone; reduces mid-radius segregation	Stirs final liquid to redistribute solute-enriched melt; reduces centreline defects
Defects Reduced	Surface cracks, uneven shell growth, inclusions near surface	Columnar grain dominance, mid-radius porosity and segregation	Centreline segregation, shrinkage cavities, V-segregation
Typical Current	200–600 A	100–400 A	50–300 A
Typical Frequency	2–8 Hz	1–5 Hz	1–3 Hz

4. Solidification During Continuous Casting

4.1 Primary Solidification in the Copper Mould

Molten steel enters the water-cooled copper mould through the SEN. The high thermal conductivity of copper enables extremely rapid heat extraction from the steel surface. At ASIL, the primary cooling water flow rate is maintained at **1800 litres per minute (LPM)**, ensuring intense and uniform heat extraction across the mould wall. Within the mould, solidification begins by heterogeneous nucleation at the mould wall, where the steel temperature drops below the liquidus almost instantly. A very thin but dense chill zone of fine equiaxed grains forms first, followed by the growth of elongated columnar dendrites oriented perpendicular to the mould wall and opposite to the direction of heat flow.

As the strand descends and more heat is extracted, the solidification front progresses inward towards the billet centre. Near the centre, dendrite fragments that detach from growing arms act as new nucleation sites, producing a zone of fine equiaxed grains. A continuously cast billet therefore displays a characteristic three-zone solidification structure: the outer chill zone, a middle columnar grain zone, and an inner equiaxed grain zone. The relative proportions of these zones are governed by casting speed, cooling intensity, steel composition, and the application of electromagnetic stirring.

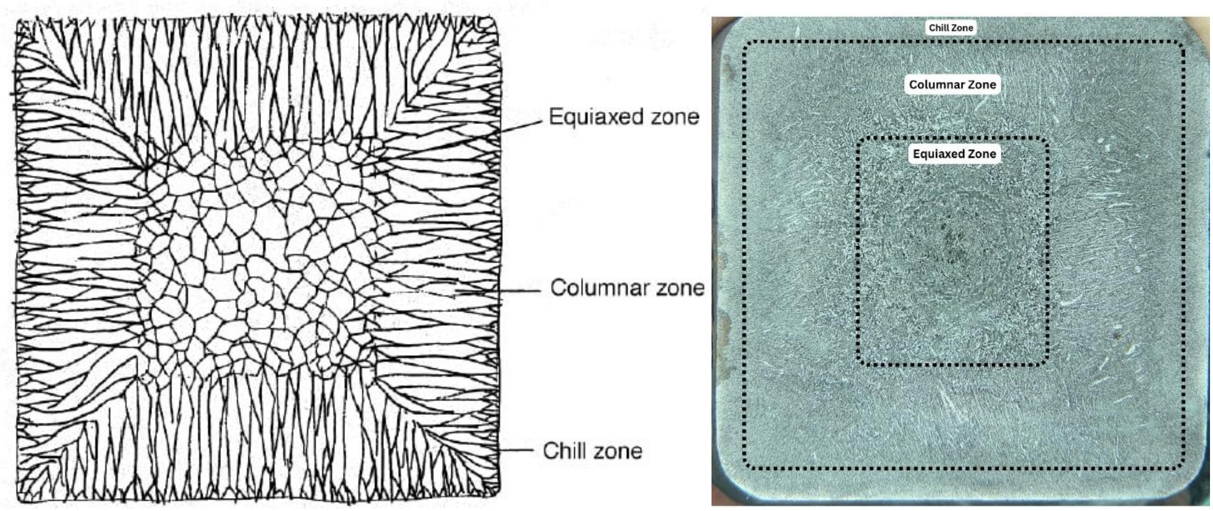

Figure 3.9: *Different types of Zones formed during Solidification at CCM*

4.2 Liquid Pool Formation

As molten steel enters the water-cooled copper mould through the SEN, heat is extracted rapidly from its outer surface, causing immediate solidification and the formation of a thin yet sufficiently strong shell capable of containing the liquid metal within. However, due to the relatively low thermal conductivity of steel and the continuous movement of the strand, the interior remains in the molten state, creating a central liquid region known as the liquid pool or liquid sump. As the strand travels downward through the mould and secondary cooling zone, continuous heat extraction causes the solidification front to progress gradually towards the centre, increasing shell thickness while reducing the size of the liquid pool until complete solidification occurs at the end of the metallurgical length.

The dimensions and depth of the liquid pool are influenced by casting speed, billet cross-section, steel composition, superheat, and the intensity of primary and secondary cooling. An increase in casting speed or excessive superheat delays solidification, resulting in a deeper liquid pool and a longer metallurgical length, whereas efficient cooling accelerates heat removal and shortens the liquid pool. During the final stages of solidification, the remaining liquid becomes enriched with alloying and impurity elements rejected by the growing dendrites, making this region highly susceptible to centerline segregation and shrinkage porosity.

4.3 Centerline Segregation and Solidification Shrinkage

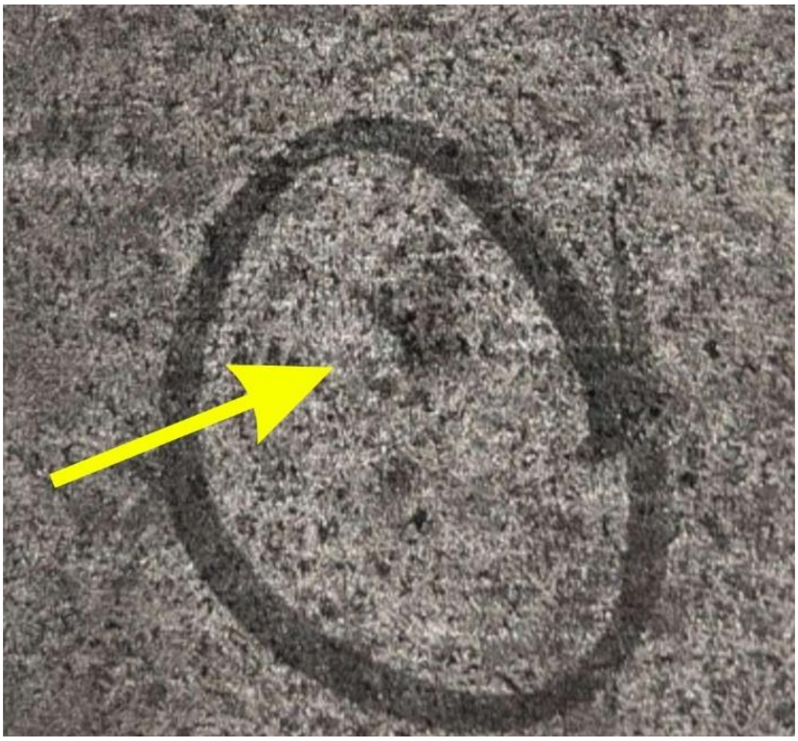

Figure 3.10: *Centerline Segregation*

During dendritic solidification, alloying elements such as carbon, manganese, phosphorus, and sulphur are rejected by the growing solid dendrites due to their partition coefficients and become progressively enriched in the remaining liquid. The final liquid to solidify is therefore highly concentrated in these solutes. If this enriched liquid accumulates and solidifies at the billet centre, it produces a centerline segregation band a region of non-uniform chemical composition that can adversely affect hardness uniformity, fatigue resistance, and machinability of the finished product. In parallel, steel undergoes volumetric contraction during solidification, which if insufficient liquid steel supply is available leads to centerline porosity and shrinkage cavities. Both segregation and porosity must be actively managed through proper control of casting conditions and the use of electromagnetic stirring.

4.4 Secondary Cooling Zone

After the strand exits the copper mould, it enters the secondary cooling zone where it is supported by a series of driven rollers. Water sprays directed at the strand surfaces extract heat and progressively complete the solidification of the billet core. The cooling intensity in this zone is carefully controlled according to the steel grade, billet cross-section, casting speed, and the superheat of the steel in the tundish. Too rapid cooling can induce thermal stresses leading to surface and internal cracks, while insufficient cooling prolongs the liquid core and risks bulging or breakout. Proper secondary cooling is therefore essential to producing a defect-free billet with sound internal structure.

5. Tundish Level and Temperature Control

During continuous casting, maintaining a stable tundish level and an appropriate steel temperature is essential for ensuring smooth metal flow, stable mould operation, and consistent billet quality. The weight of liquid steel in the tundish is continuously monitored using load cells mounted beneath the tundish support structure. Maintaining a constant tundish level ensures a stable ferrostatic head acting on the SEN, providing uniform flow of molten steel into the mould. Fluctuations in the tundish level can alter the steel flow rate, disturb the mould level, and increase the likelihood of defects such as slag entrapment and surface irregularities.

The temperature of molten steel in the tundish is periodically measured throughout the casting process to ensure it remains within the specified operating range. The key parameter used for temperature control is superheat — defined as the difference between the casting temperature of molten steel and its liquidus temperature: Superheat = T(Casting) − T(Liquidus). Maintaining an optimum superheat is crucial for stable casting operations. If superheat is too low, premature solidification may occur inside the SEN or mould, leading to flow instability and nozzle clogging. Conversely, excessive superheat increases the depth of the liquid pool, delays complete solidification, and promotes centreline segregation and internal porosity.

The liquidus temperature used in the superheat calculation is not a fixed value it varies with every heat depending on the actual steel chemistry. At ASIL, it is calculated using the following empirical formula displayed on the shop floor:

$$\mathbf{T_{Liquidus} = 1539 - (59C + 22Al + 31(S+P) + 11.5\,Si + 3.6\,Mn + 3.8\,Ni + 1.8\,Cr + 2.3\,Mo + 4.3\,Cu)}$$

Where each element symbol represents its weight percentage in the steel

Each alloying element lowers the liquidus temperature by a specific amount. Carbon has the strongest depressing effect at 59°C per weight percent, which is why high carbon steels have a significantly lower liquidus temperature than low carbon steels and must be handled differently during casting. The operator recalculates the liquidus temperature for every heat using the final chemistry from the LRF sample, and uses it to determine the correct tundish casting temperature and confirm whether the steel has adequate superheat before the sequence begins.

The metallurgical length is defined as the distance from the meniscus of molten steel in the mould to the point where complete solidification of the strand occurs, representing the length of the liquid core inside the continuously cast billet. An increase in casting speed or superheat increases the metallurgical length, while enhanced cooling rates reduce it. A properly controlled metallurgical length is essential for producing billets with good internal quality — an excessively long liquid pool promotes centreline segregation and shrinkage defects, whereas an excessively short metallurgical length may lead to high thermal stresses and surface cracking.

6. Key Process Parameters Monitored During Casting

The quality and consistency of continuously cast billets depend on the precise control of several process parameters throughout the casting sequence. These parameters influence the solidification behaviour, internal soundness, surface quality, and dimensional accuracy of the billets. The major process parameters monitored during casting are:

- SEN (Submerged Entry Nozzle) immersion depth
- Mould steel level
- Casting speed
- Primary cooling water flow rate and pressure
- Secondary cooling water flow rate
- Electromagnetic Stirrer (EMS) current and frequency
- Tundish steel temperature and superheat
- Argon gas pressure and flow rate
- Billet cross-sectional dimensions
- Mould oscillation frequency and stroke

Any deviation of these parameters from the prescribed operating range can adversely affect the casting process, leading to defects such as surface cracks, centerline segregation, porosity, slag entrapment, breakout, or dimensional variations.

7. Billet Withdrawal and Cutting

Figure 3.11: *Billet withdrawing*

The solidified strand is continuously withdrawn from the mould by a set of driven withdrawal rollers that maintain the casting speed at the set value. As the strand descends and passes through the secondary cooling zone, solidification is completed and the billet reaches full cross-sectional solidity. The solid billet is then cut to the required length using an oxy-fuel cutting torch that moves synchronously with the strand. As a matter of standard practice, the first and last billets of each casting sequence are discarded, since these transition pieces are likely to have quality variations arising from the non-steady-state conditions at the start and end of the heat.

At Aarti Steel International Limited, the CCM is capable of producing five standard billet cross-sections depending on the product requirement and customer order:

- **120 × 120 mm:** smallest section; used for lighter bar and rod rolling
- **140 × 140 mm:** intermediate section for general engineering steel products
- **160 × 160 mm:** medium section for structural and alloy steel applications
- **200 × 200 mm:** larger section for heavy engineering and forging grade billets
- **240 × 240 mm:** largest section; used for heavy forging and special alloy steel requirements.

7.1 Casting Speed and Its Metallurgical Basis

Casting speed is one of the most critical parameters in continuous casting and is not fixed uniformly for all steel grades. At ASIL, the casting speed varies depending on the carbon content and grade of steel being cast, as the carbon content directly influences the solidification behaviour, heat generation, and shell growth rate of the strand.

- For **low carbon steels**, a relatively higher casting speed can be maintained because low carbon steel has a wider solidification range and generates less heat during solidification, allowing the shell to grow more uniformly and withstand the ferrostatic pressure of the liquid core without the risk of breakout.
- For **high carbon steels**, the casting speed is deliberately kept lower. High carbon steel has a narrower solidification range and releases significantly more heat during solidification. This results in a thinner and more brittle shell at the mould exit, making it highly susceptible to longitudinal cracking and breakout if the speed is too high. A lower casting speed allows more time for the shell to thicken sufficiently before it exits the mould and enters the secondary cooling zone.

8. Key Process Parameters Monitored During Casting

The quality and consistency of continuously cast billets depend on the precise control of several process parameters throughout the casting sequence. These parameters influence the solidification behaviour, internal soundness, surface quality, and dimensional accuracy of the billets. Continuous monitoring and timely adjustment of these variables are essential to ensure stable casting conditions and to prevent the formation of casting defects.

The major process parameters monitored during casting include:

- SEN (Submerged Entry Nozzle) immersion depth
- Mould steel level

- Casting speed
- Primary cooling water flow rate and pressure
- Secondary cooling water flow rate
- Electromagnetic Stirrer (EMS) current and frequency
- Tundish steel temperature and superheat
- Argon gas pressure and flow rate
- Billet cross-sectional dimensions
- Mould oscillation frequency and stroke

Any deviation of these parameters from the prescribed operating range can adversely affect the casting process, leading to defects such as surface cracks, centerline segregation, porosity, slag entrapment, breakout, or dimensional variations. Therefore, continuous monitoring and process control are essential for producing billets with consistent quality and sound internal structure.

9. Conclusion

The Continuous Casting Machine integrates an intricate sequence of metallurgical, mechanical, and process control operations that must function in unison to convert liquid steel reliably into high-quality solid billets. Beginning with disciplined tundish preparation, cleaning, refractory coating, controlled preheating, and secure SEN installation and continuing through carefully managed steel flow, effective mould lubrication with casting powder, precise mould level control, optimised secondary cooling, and electromagnetic stirring, every stage of the process contributes to the internal soundness and surface integrity of the final billet.

End of Internship Report